



**Cyber-Informed  
Engineering**

# **Requirements Framework for Cyber-Informed Engineering**

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**January 2025**  
**NREL/TP-5T00-90317**

Cyber-Informed Engineering (CIE) Program activities are sponsored by the U.S. Department of Energy's Office of Cybersecurity, Energy Security, and Emergency Response (DOE CESER) and performed by Idaho National Laboratory and the National Renewable Energy Laboratory.

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## Executive Summary

In an era where cyber threats are increasingly sophisticated and pervasive, traditional engineering practices must evolve to integrate cybersecurity considerations from the outset. Cyber Informed Engineering (CIE) addresses this need by embedding cybersecurity principles into the engineering life cycle, ensuring that systems are designed to be secure and resilient against potential cyber-attacks.

This paper aims to develop a comprehensive requirements framework for CIE, detailing how traditional requirements engineering processes can be customized to incorporate key CIE principles. The goal is to provide a structured approach that organizations can use to ensure that cybersecurity is an integral part of their system design and implementation processes. The framework is developed through an in-depth exploration of the requirements engineering process, including requirements elicitation, specification, validation, and management. Each stage is tailored to address cybersecurity concerns by integrating activities such as formal reviews, prototyping, and model checking. Specific techniques for eliciting, writing, modeling, reviewing, and validating requirements are detailed, with a focus on quantifying the feasibility of implementing CIE principles.

By integrating cybersecurity considerations into the requirements engineering process, the proposed framework ensures that security is not an afterthought but a fundamental component of system design. This approach enhances the resilience and security of engineered systems, helping organizations proactively mitigate cyber threats and reduce ambiguity. The framework serves as a valuable tool for engineers and cybersecurity professionals, providing a systematic way to embed robust security practices into the engineering life cycle.

## Acknowledgments

This work was authored by the National Renewable Energy Laboratory, operated by Alliance for Sustainable Energy LLC, for the U.S. Department of Energy under Contract No. DE-AC36-08GO28308, and by Idaho National Laboratory, operated by Battelle Energy Alliance LLC, for the U.S. Department of Energy under DOE Idaho Operations Office Contract DE-AC07-05ID14517. Funding provided by the U.S. Department of Energy Office of Cybersecurity, Energy Security, and Emergency Response. The views expressed in the article do not necessarily represent the views of the DOE or the U.S. Government. The U.S. Government retains and the publisher, by accepting the article for publication, acknowledges that the U.S. Government retains a nonexclusive, paid-up, irrevocable, worldwide license to publish or reproduce the published form of this work, or allow others to do so, for U.S. Government purposes.

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# 1. Introduction

In today's interconnected digital landscape, cybersecurity is not just a technical necessity but a foundational element of modern engineering. The frequency and sophistication of cyber-attacks are increasing, posing significant threats to critical infrastructure, enterprises, and individual users. Traditional engineering practices often consider security as an add-on, leading to systems that are inherently vulnerable to cyber threats. Cyber Informed Engineering (CIE) is a transformative approach that mitigates cybersecurity challenges through engineering means throughout the system engineering life cycle, ensuring that systems are designed with security and resilience at their core.

The need for a proactive approach to cybersecurity has never been more urgent. Cyber-attacks can have devastating consequences, from financial losses and operational disruptions to the erosion of customer trust and effects on national security. Despite the growing recognition of these risks, many organizations struggle to integrate cybersecurity effectively into their engineering processes.

While existing guides such as CIE implementation guide integrate cybersecurity into the engineering process, there remains a gap in the practical implementation of CIE principles. Many engineering teams lack a structured framework to guide the incorporation of CIE principles throughout the development life cycle. This gap can contribute to systems that are less resilient to evolving cyber threats, potentially increasing the risk of financial losses, compromised data integrity, and operational disruptions. Addressing this gap is critical for enhancing the security and resilience of engineered systems.

The purpose of this paper is to provide a detailed and actionable requirements framework for incorporating CIE principles in the system engineering lifecycle. By systematically embedding cybersecurity principles into each stage of the requirements engineering process, this framework aims to bridge the gap between traditional engineering practices and the demands of modern cybersecurity. The impact of this document is far-reaching, offering a practical guide for organizations across various industries to enhance their security posture and mitigate the risks posed by cyber threats.

The methodology employed in this paper involves in-depth customization of the traditional requirements engineering process to incorporate CIE principles. This includes detailed activities for requirements elicitation, specification, validation, and management, each tailored to address specific cybersecurity concerns. Techniques such as formal reviews, prototyping, and model checking are employed to ensure that the requirements are both practical and effective. Quantifiable metrics are introduced to assess the feasibility of implementing CIE principles, providing a clear and measurable framework for organizations to follow.

Chapter 2 describes requirement engineering process, requirement models used, and background about cyber-informed engineering. Chapter 3 describes the intended audience for this framework. Chapter 4 describes the requirements framework for cyber-informed engineering. Finally, Chapter 5 concludes the requirement framework paper.

## 2. Background

This section provides background on the requirements engineering process, requirement models in literature, and an overview of CIE. Readers familiar with requirement processes and CIE may consider skipping this section.

### 2.1. Requirements Engineering Stages

Requirements engineering involves several interconnected steps, each critical to the successful development of a project.

#### 2.1.1 REQUIREMENTS ELICITATION

The goal of this step is to gather comprehensive information from stakeholders to understand their needs and constraints. This process is achieved by several approaches such as:

- Interviews: The interviews can be either structured or unstructured. The structured interviews involve predefined questions to ensure all aspects are covered and unstructured interviews involve open-ended discussions to explore broader requirements.
- Surveys/Questionnaires: The surveys/questionnaires are useful for broader reach to collect data from a larger number of users.
- Workshops: The workshops involve collaborative sessions for stakeholders to discuss and brainstorm requirements.
- Observation: The observation involves contextual inquiry to observe users in their environment.
- Document analysis: The document analysis involves reviewing the current systems, manuals, and business processes.

#### 2.1.2 REQUIREMENTS SCOPING AND DECONFLICTION

The goal of this step is to scope, prioritize, and resolve conflicts among gathered requirements. This process is achieved by

- Classifying requirements: The classification of requirements involves categorizing needs based on business requirements [1], user/stakeholder requirements[2], architectural requirements [3], structural requirements [4], behavioral requirements [5], functional requirements (solution) [6], non-functional requirements (quality of service) [7], implementation requirements [8], and regulatory requirements [9].
- Prioritization: The prioritization involves ranking [10] based on MoSCoW method [11] of Must have, Should have, Could have, and Won't have this time. Some of the other prioritization techniques are Ranking, Numerical Assignments (Grouping), Bubbles Sort Techniques, Hundred Dollar Method, Analytic Hierarchy Process (AHP), Kano Analysis, Five Whys, Prioritization Based on Value, Cost, and Risk, Pairwise comparison
- Conflict resolution: The conflict resolution involves reconciling potentially incompatible requirements through stakeholder discussions [12].



### 2.1.3 REQUIREMENTS SPECIFICATION

The goal of this step is to document the requirements in a clear and detailed manner to ensure they are understood and agreed upon by all stakeholders. This process is achieved by

- Writing requirements: The requirements should be written in a clean and precise language avoiding ambiguity [13] [14] [15]. Requirements should also be capable of standing alone and not combined such that they can be validated independently.
- Modeling requirements: The requirements can be modeled using various methods, like creating a use case diagram that visually represents the system interaction or a data flow diagram that illustrates data movement within the system [16] [17].
- Reviewing and validating: This involves regular stakeholders reviews to ensure requirements are correctly captured [18].

### 2.1.4 REQUIREMENTS VALIDATION

The goal of this step is to ensure that the documented requirements are accurate, complete, and feasible. This process is achieved by

- Formal reviews: This involves step-by-step walk through review of the requirements and inspections through detailed examination of the requirements document.
- Prototyping: The prototyping involves either high-fidelity prototypes that are interactive or low-fidelity prototypes that involves simple sketches [19].
- Model checking: The model checking is a formal mathematical technique to ensure that requirements are logically sound and implementable [20].

### 2.1.5 REQUIREMENTS MANAGEMENT

The goal of this step is to manage changes to the requirements and maintain their consistency throughout the project life cycle. This process is achieved by

- Change management: The change management involves processes such as formal change requests for proposing changes and impact analysis to assess the implications of changes [21].
- Traceability: The traceability involves using a requirements traceability matrix (RTM) [22] to link requirements to their origins and implementation.
- Version control: Version control involves documenting versions to maintain versions of the requirement document [23].

## 2.2. Requirements Engineering Models

There are many engineering models described in literature. Some of them such as a pure linear model, linear with iterations between activities model, iterative requirement model, and finally an effective requirement engineering process are described. These models are helpful in understanding the steps in the requirements process.

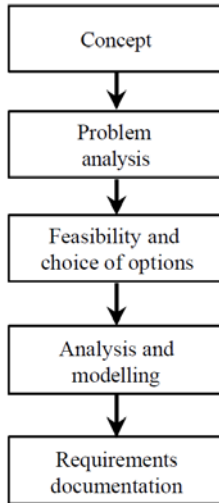


Figure 1 Pure linear model [24]

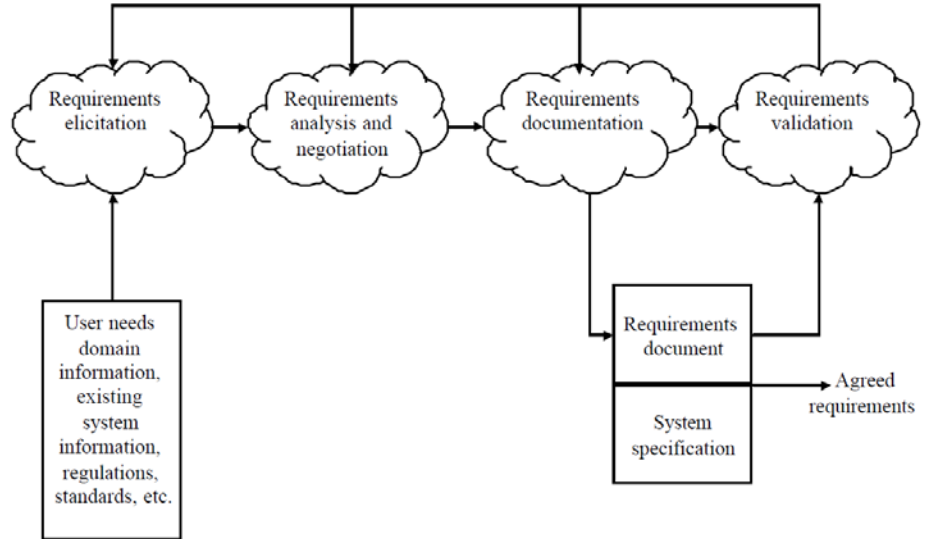


Figure 2 Linear with iterations between activities model [25]

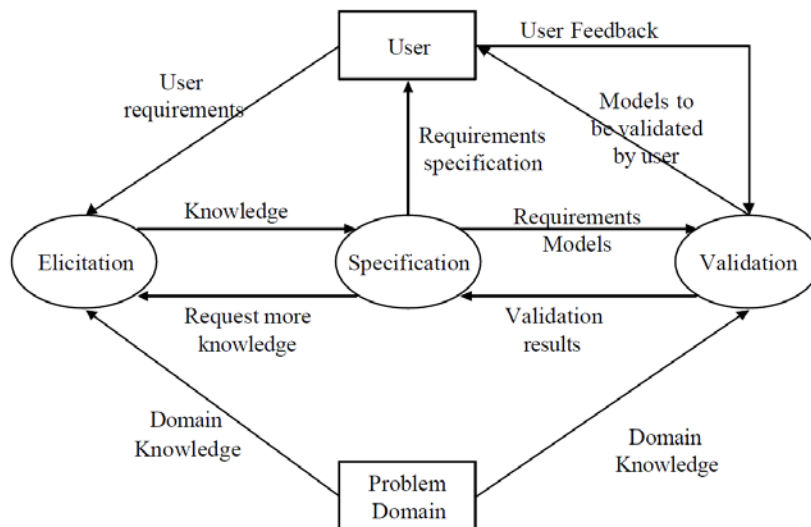


Figure 3 Iterative requirement model [26]

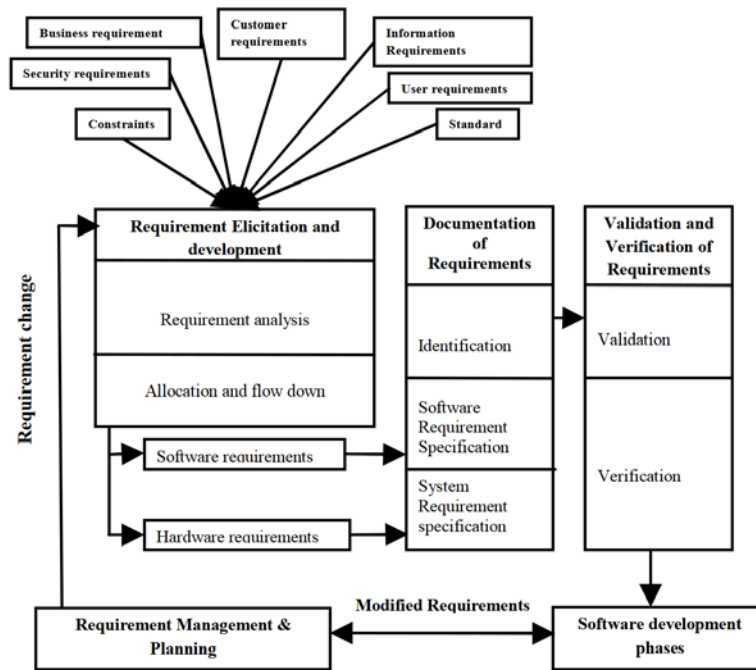


Figure 4 Effective requirement engineering process model [27]

Table 1 Summary of the models

| Model                                           | Description                                                                                                                                                                                                                                                                                                                                                                             |
|-------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Pure Linear Model                               | The pure linear model is shown in Figure 1. The pure linear model is a sequential approach where each phase must be completed before the next step begins. This model is straightforward and easy to manage, making it suitable for projects with well-defined requirements that are unlikely to change.                                                                                |
| Linear with iterations between activities model | The linear with iterations between activities model is shown in Figure 2. This model develops the system through repeated cycles (iterative) and in smaller portions at a time (incremental). It allows for partial implementation and subsequent refinement, making it easier to adapt to changes in requirements.                                                                     |
| Iterative requirement model                     | The iterative requirement model is shown in Figure 3. This model is a flexible, iterative approach that promotes continuous feedback and adaptive planning. It emphasizes collaboration and customer satisfaction through the delivery of small, functional increments of the product.                                                                                                  |
| Effective requirement engineering process model | The effective requirement engineering process model is shown in Figure 4. This model is a robust requirements engineering model aimed at generating high-quality requirements for software development. Emphasizing independent requirement management and planning phases, it adopts an iterative approach to enhance both initial requirement engineering and subsequent maintenance. |

## 2.3. Cyber-Informed Engineering

As per the national cyber-informed engineering strategy outlined in [28], CIE is an emerging framework that integrates security controls directly into the design of systems from the very

beginning. It combines both the physics and mechanics of engineering with digital protections to enhance the security of the nation's energy systems and other critical infrastructures. This approach targets systems with digital connectivity, monitoring, or process control, ensuring they are secure from the earliest stages of development.

Unlike traditional methods, which often focus on securing systems after deployment, CIE emphasizes proactive security integration during the initial design phase. This proactive approach ensures that security is inherent in the system's foundation, rather than an afterthought. CIE achieves this by leveraging design choices and engineering controls to either eliminate or mitigate vulnerabilities that could be exploited in cyberattacks.

CIE principles contribute to determining the critical functions that the system delivers and preventing undesirable outcomes through consequence-driven design, using engineering controls to minimize vulnerabilities, safeguarding the integrity of vital data with secure information architecture, streamlining the design by removing unnecessary elements to achieve design simplification, establishing a robust set of system defenses to ensure resilient layered defenses, and proactively defending the system against threats through active defense.

Additionally, the system can have an impact on other systems or be impacted by them, which requires inter-dependency evaluation. Gaining a clear understanding of the presence and functioning of digital assets ensures digital asset awareness, while ensuring that supply chain providers adhere to security standards supports cyber-secure supply chain control. Converting 'what ifs' into 'even ifs' fosters planned resilience, safeguarding the knowledge of the system to prevent it from falling into the wrong hands through engineering information control, and ensuring that everyone's behavior and decisions aligns with the overarching security goals promotes a cybersecurity-aware culture.

### **3. Intended Audience**

The requirements framework for CIE is designed to assist and enhance the existing system engineering lifecycle, not to create a new methodology for developing requirements. CIE leverages established engineering processes and frameworks, integrating its principles to help engineers and stakeholders think critically about the digital consequences of their designs and systems. By incorporating CIE, the focus is on improving the quality of requirements through a consequence-driven approach, ensuring that engineering solutions proactively mitigate potential digital vulnerabilities. This way, CIE supports and strengthens the traditional requirements process without reinventing it. The intended audience for the framework can be classified into primary and secondary audience.

#### **3.1 Primary Audience**

The audience who will use this framework throughout the system engineering life cycle for developing a system that is secure based on the CIE principles.

- **System Engineers and Developers:** Professionals involved in designing and building systems who need to integrate cybersecurity measures throughout the development process

- **Project Managers:** Individuals overseeing system development projects who need to ensure that cybersecurity is a prioritized and integrated component of the project life cycle.
- **Quality Assurance and Testing Teams:** Professionals tasked with verifying that the implemented system meets the specified security requirements and is resilient to cyber threats.

## 3.2 Secondary Audience

The audience who will be consulted by the primary audience to successfully build the system throughout the engineering life cycle in accordance with CIE principles.

- **Cybersecurity Experts:** Specialists responsible for identifying, analyzing, and mitigating cyber threats who can provide insights and validation for security requirements.
- **Policy makers, Regulatory Officials, and Compliance Officers:** Personnel responsible for ensuring that systems meet regulatory, and compliance standards related to cybersecurity.
- **Business Stakeholders and Executives:** Decision-makers who need to understand the importance of embedding cybersecurity into engineering practices to protect organizational assets and maintain customer trust.
- **Academic and Research Communities:** Scholars and researchers studying cybersecurity and engineering methodologies who can benefit from a structured framework for integrating cybersecurity principles into system design and development.

## 4. Requirement Framework for CIE

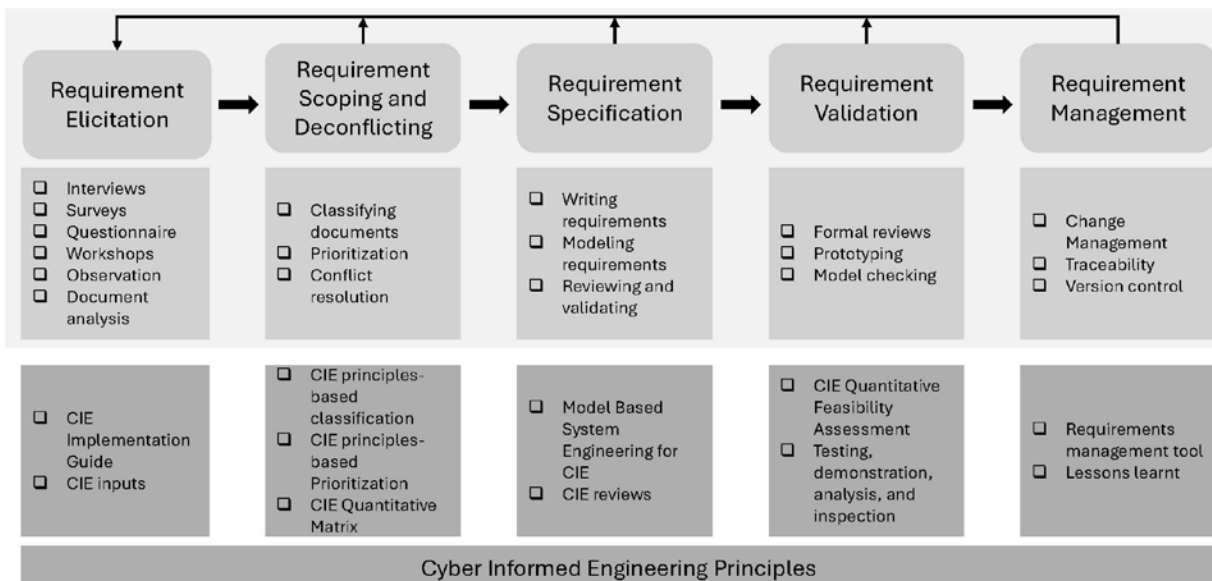


Figure 5 Requirement framework for CIE

The requirement framework for CIE is shown in Figure 5. The elicitation, scoping, and deconflicting requirements are initial steps in forming requirements. The requirements specifications are the actual requirements that will be used in the system engineering life cycle. The requirements validation and management are used throughout the system engineering life cycle to develop the system according to the requirements specified and log all the requirement changes.

## 4.1. Requirements Elicitation

When using CIE, the requirements elicitation phase is critical to ensure that we look at a requirement in the round and tease out all the needs for engineering-based protections and layered defenses. This stage ensures that all relevant security concerns are identified and considered from the outset, forming the foundation for a secure system design. The elicitation process involves multiple techniques, including interviews, surveys/questionnaires, observations, and document analysis. Each technique provides unique insights and complements the others to create a comprehensive understanding of the cybersecurity requirements. Most of the questions here are for examples and a comprehensive list is in the CIE implementation guide [29].

### 4.1.1 INTERVIEWS

Interviews are a direct and effective way to gather detailed information from stakeholders. For CIE, interviews should focus on understanding the broader operational and engineering protections necessary to mitigate the potential for cyber sabotage. The stakeholders to be interviewed include business owners, the operational team, engineers, IT personnel, cybersecurity experts, and end-users. The key questions to ask might include the following.

- Business Owners: What are the most critical assets and functions that need protection? What are the potential operational consequences of a cyber-attack on these assets? How do you currently ensure resilience against system disruptions?
- Operational Teams: What operational processes and functions are most susceptible to disruption from external threats? How do you currently manage resilience and continuity in critical operations? What impact would a failure in engineering-based defenses have on system performance?
- Engineers: What engineering controls are in place to prevent system disruptions and failures? How can these controls be strengthened to minimize the risk of cyber sabotage? Are there specific vulnerabilities in the system's design that could be exploited?
- IT Staff: What security protocols and defenses are integrated with engineering systems? What vulnerabilities in the current system could impact critical functions? How do you handle incidents that can affect operational technology (OT)?
- Cybersecurity Experts: What are the latest threats and attack vectors relevant to our industry, particularly those that target operational technologies? How do we align cybersecurity measures with engineering protections to prevent system sabotage? What best practices can improve resilience through the design of the system?

Including these broader perspectives ensures that interviews address both operational and engineering needs, capturing the requirements necessary to build protection from cyber sabotage through CIE principles rather than relying solely on traditional IT security measures.

#### 4.1.2 SURVEYS/QUESTIONNAIRES

Surveys and questionnaires are useful for gathering information from a larger group of stakeholders, providing a broader perspective on cybersecurity requirements. These tools can be used to collect quantitative and qualitative data on security priorities, user experiences, and perceived risks. Questions might include:

- What are the systems that perform and support critical facility functions?
- What consequences of failure or maloperation are the engineered controls designed to prevent?
- What engineering-based protection and verification could ensure that key data elements have not been manipulated?
- What layers of engineered control defenses exist in a system and assess their interdependence from other defenses and area of effect?
- What temporary operational changes can be made in response to a perceived threat?
- What outputs does the system provide that are critical inputs to other business systems or infrastructures?

Surveys can also include scenario-based questions to gauge stakeholder responses to potential security incidents. This helps in understanding stakeholder expectations and preparedness for various security scenarios.

#### 4.1.3 OBSERVATIONS

Observing stakeholders in their natural environment provides valuable information on how they interact with the system and can reveal potential vulnerabilities that may not be apparent through interviews or documentation. For CIE, observations should focus on how system operators interact with and manage critical assets, how engineers and operational staff implement and maintain engineering-based protections, and how routine tasks are performed in a real-world context to ensure system resilience. Key observations might include the following.

- How can multiple concurrent compromises of already in-place mitigations (i.e., n-2, n-x) contribute to more severe consequences?
- How are engineering controls monitored and reassessed to identify changes from evolving operations that may weaken their effectiveness?
- How undesired manipulation of important data is prevented?
- How are features of the system that are not necessary to achieve the critical functions determined?

Instead of focusing on IT staff managing security protocols, the emphasis could be on how operational controls and engineering measures are applied and adapted in daily activities to prevent potential disruptions or sabotage. This approach aligns with the CIE's goal of engineering out vulnerabilities and enhancing operational defenses.



#### 4.1.4 DOCUMENT ANALYSIS

Document analysis involves reviewing key artifacts that are integral to the system engineering lifecycle, such as system security plans, interface control documents, and system design documents. These documents provide crucial insights into the existing engineering framework, operational controls, and how security measures are integrated into system architecture. For CIE, important documents to review might include:

- System Security Plan: What security measures are in place, and how are they integrated into the system's overall design and operation? Does the plan address potential vulnerabilities engineered into the system?
- Interface Control Documents: How do different system components interact? Are there any interface vulnerabilities that could be exploited?
- System Design Documents: How is the system architected? Are there engineered redundancies or defenses that minimize the potential impact of a cyber-attack?

Analyzing these documents helps provide a complete understanding of the design, operational procedures, and protections of the system, allowing identification of areas where additional security or resilience measures may be required. This ensures alignment with CIE principles while addressing gaps in the system's overall security posture.

#### 4.1.5 INTEGRATING INSIGHTS

Integrating insights from interviews, surveys, observations, and document analysis is a critical and complex step in the requirements elicitation process. This process involves not only collecting data, but also synthesizing and analyzing it to ensure that all perspectives, operational, engineering, and cybersecurity, are represented in the final set of requirements. To do this effectively, it is important to develop a structured approach that focuses on categorizing and correlating the information gathered from different sources.

An effective strategy is to use a thematic analysis, where the data is grouped into themes such as asset protection, operational reliability, supply chain integrity, and built-in defenses. By identifying recurring themes, the team can ensure that the system's most critical components and functions are protected by appropriate engineering and security measures. For example, interviews with system operators might reveal key insights into operational dependencies, while document analysis could uncover architectural vulnerabilities that need to be addressed.

Cross-functional workshops can also help integrate information by bringing together stakeholders from different domains: operations, engineering, IT, and security, allowing them to collectively review the findings and validate the emerging requirements. These workshops should focus on reconciling any conflicts between operational needs and security requirements, ensuring that solutions are balanced and feasible within the constraints of the system. Finally, prioritizing requirements based on their alignment with CIE principles helps to ensure that the most critical aspects of the system are addressed first, supporting both resilience and security from the start of the design process.

This structured, multi-perspective approach enables a more complete and cohesive set of requirements, ensuring that security is embedded deeply into the system's architecture and operations, and not treated as an afterthought.



## 4.2. Requirements Scoping and Deconfliction

When using CIE, the requirements scoping and deconfliction stage is pivotal. It involves dissecting the gathered requirements to understand their implications, prioritizing them based on various criteria, and resolving any conflicts that may arise between different stakeholder needs. This stage ensures that cybersecurity requirements are not only comprehensive, but also feasible and aligned with the overall goals of the system.

### 4.2.1 DETAILED ANALYSIS

The first step in this stage is to thoroughly analyze the requirements elicited. This involves categorizing requirements into functional (requirements that directly impact key system functions) and non-functional (requirements related to resilience, supply chain integrity, or cultural considerations) requirements. For example, functional requirements might include the need for secure information flow between critical systems in a hydroelectric power plant, or the engineering controls to ensure the integrity of turbine control systems under various operational conditions. Non-functional requirements could include the system's resilience to supply chain disruptions, the ability to maintain functionality even if external digital assets are compromised or establishing a cybersecurity-aware culture within operational staff to handle abnormal situations.

Beyond functional and non-functional classifications, it is also important to assess the digital impact of each requirement. For example, the requirement “*Fully automated control system with remote monitoring*” has direct digital consequences as it involves automation and remote monitoring, both of which are vulnerable to cyber threats. The requirement “*Sediment management system to prevent reservoir capacity loss*” has indirect digital consequences since it is not directly related to power generation but still involves digital components that may need protection. Finally, the requirement “*Dam structure to withstand a B-year flood event*” has no digital consequences, as it is purely an engineering requirement focused on the physical resilience of the dam structure.

CIE focuses on addressing both direct and indirect digital consequences by incorporating engineering mitigations to prevent incidents arising from vulnerabilities within these domains. Requirements with direct digital consequences, such as those involving automation or digital monitoring, and those with indirect digital components, such as systems that interface with digital technologies, are within the scope of CIE. However, requirements with no digital consequences, such as purely physical engineering concerns, fall outside the scope of CIE.

### 4.2.2 PRIORITIZATION AND DECONFLICTING

Once the requirements are analyzed, the next step is to prioritize them. Prioritization is essential in scenarios where resources are limited, and not all requirements can be implemented immediately. One effective method for prioritizing cybersecurity requirements is the MoSCoW method, which categorizes requirements into Must have, Should have, Could have, and Won't have this time. This helps in focusing on the most critical security measures first.

During prioritization, it's important to consider the input from all stakeholders, including business owners, operational team, engineers, IT staff, and cybersecurity experts. This often involves

negotiating to resolve conflicts between different stakeholder needs. For example, business owners might prioritize features that enhance user experience, while IT staff may focus on robust security measures that could potentially complicate the user experience. Negotiation sessions should aim to find a balance that ensures the system is secure without compromising on usability or other critical business functions.

When applying CIE, it is crucial to focus on requirements that address digital consequences, such as remote monitoring, automated control systems, or interactions with critical infrastructure. The prioritization process should be driven by the potential impact of digital failure or sabotage on key operational functions, rather than just traditional cybersecurity threats. Requirements that protect critical operations, where a digital compromise could lead to severe safety or operational impacts, should take precedence.

CIE-driven prioritization evaluates both the likelihood and impact of digital incidents, aligning engineering mitigations with the most significant consequences. This ensures that resources are focused where failure would have the greatest impact. By designing resilient systems, CIE helps reduce or eliminate digital risks, creating more secure systems better equipped to handle both direct and indirect digital consequences.

#### 4.2.3 HYPOTHETICAL CIE QUANTITATIVE MATRIX

To systematically prioritize cybersecurity requirements and align them with CIE principles, we can develop a CIE quantitative matrix. One such matrix could be to map the requirements against key CIE principles, allowing for a structured and quantitative approach to prioritization. The matrix could consist of the following steps:

- **Identify Key CIE Principles:** List the essential CIE principles such as consequence-driven design, engineering controls, secure information architecture, design simplification, resilient layered defenses, active defense, inter-dependency evaluation, digital asset awareness, cyber-secure supply chain control, planned resilience, engineering information control, and fostering a cybersecurity culture.
- **List the Requirements:** Enumerate all the requirements with direct and indirect digital consequences identified during the analysis stage.
- **Score the Requirements against CIE Principles:** For each requirement, score its importance or impact with respect to each CIE principle on a predefined scale. This score represents how well the CIE principles are included during the requirement phase (e.g., 1-5, where 1 is did not follow CIE principle and 5 is followed all/most of the CIE principles).
- **Prioritize Requirements:** Rank the requirements based on their total scores. Higher scores indicate requirements that better align with CIE principles and lower scores indicate requirements did not follow CIE principles. This gives us opportunity to prioritized requirements for further analysis to better align with the CIE principles.

### 4.3. Requirements Specification

The requirements specification stage is a detailed process that transforms elicited and analyzed requirements into a comprehensive and precise documentation format. This stage involves writing clear and unambiguous requirements, modeling these requirements to provide a visual

representation of system interactions and behaviors, and thoroughly reviewing and validating these specifications to ensure accuracy and completeness. The depth of modeling required is crucial for effectively implementing CIE principles, ensuring that cybersecurity measures are both robust and integrative.

#### 4.3.1 WRITING REQUIREMENTS

When using CIE, writing requirements necessitates precision and clarity to avoid ambiguity and ensure that all stakeholders have a shared understanding of what needs to be achieved. Requirements must be articulated in a way that leaves no room for misinterpretation, specifying not only the functional aspects but also the security-related non-functional requirements.

For instance, when specifying a requirement for digital asset tracking within a secure information architecture, it should detail the processes for identifying and cataloging critical assets, the mechanisms for continuous monitoring of asset status and integrity, and the procedures for responding to unauthorized access or changes. This level of detail ensures that engineers and system administrators can implement and verify the requirement effectively, minimizing the need for further clarification.

Moreover, CIE requirements should explicitly address key CIE principles. For example, a requirement might specify that the system must be able to detect and respond to unauthorized access attempts within a defined time frame, reflecting the principle of active defense. By writing requirements that directly reference CIE principles such as consequence-driven design, secure information architecture, and resilient layered defenses, the specification document becomes a comprehensive guide for building secure systems.

#### 4.3.2 MODELING REQUIREMENTS

Modeling requirements involves creating detailed visual representations that capture interactions, data flows, and security controls within a system, with the complexity and depth of modeling tailored to the system's criticality and security features. From a CIE perspective, Model-Based Systems Engineering (MBSE) offers significant advantages by providing a structured and integrated approach to capturing these requirements, ensuring that security considerations are embedded throughout the system's life cycle. MBSE allows for the precise visualization of how different components interact, where data flows, and how security controls are applied, making it easier to identify and mitigate potential vulnerabilities.

In practice, MBSE techniques like use case diagrams, data flow diagrams (DFDs), and state diagrams are instrumental in this process. Use case diagrams illustrate how users interact with the system, highlighting where security controls are necessary to protect against potential threats. DFDs are particularly valuable for visualizing the movement of data within the system, pinpointing potential points of compromise. For example, in a healthcare application, a DFD could demonstrate the flow of patient data, revealing where encryption and access controls are needed to secure sensitive information. State diagrams are also essential for modeling the system's different states and transitions, ensuring that security requirements are maintained throughout all possible states. A state diagram for a smart home system might depict states like idle, active, and alarm, with security checks such as device authentication occurring during transitions.

The use of MBSE not only facilitates a deeper understanding of the system's security architecture but also enhances the ability to systematically integrate CIE principles into the design. By providing a visual and interconnected view of the system, MBSE helps in identifying and addressing security gaps that might not be evident from textual requirements alone. This thorough and integrated modeling process ensures that security considerations are not an afterthought but a fundamental part of the system's design, providing a robust foundation for secure implementation.

### 4.3.3 REVIEWING AND VALIDATING

The review and validation process are essential to ensure that the specified requirements and models are both accurate and feasible. Requirements validation methods should be developed in parallel with the requirements development such that it ensures that the developed requirements are verifiable. Each requirement should have a validation statement for how it will be validated such as Testing, Demonstration, Analysis, and Inspection.

Validation ensures that the requirements meet the needs and expectations of all stakeholders and that they can be practically implemented. This process often involves scenarios or walkthroughs where stakeholders simulate how the system would function with the specified requirements. For instance, a validation exercise might involve testing a prototype system's response to a simulated cyber-attack, ensuring that the specified active defense mechanisms are effective.

Reviewing involves a critical examination of the requirements document and models by stakeholders, including cybersecurity experts, developers, and business representatives. This collaborative review helps in identifying any ambiguities, inconsistencies, or gaps in the requirements.

In conclusion, the requirements specification stage is a detailed and iterative process that involves precise writing, comprehensive modeling, and rigorous reviewing and validating. The depth of modeling required is critical to ensuring that CIE principles are thoroughly integrated into the system design. By achieving the right balance between detail and practicality, organizations can create robust requirements that form the foundation for secure and resilient systems.

## 4.4. Requirements Validation

When using CIE, the requirements validation stage is essential for ensuring that cybersecurity requirements are accurate, complete, and feasible within the project's constraints. This stage not only verifies that the CIE principles embedded in the requirements can be effectively implemented but also ensures that the requirements are crafted to facilitate both validation and verification processes in parallel. Effective validation involves methods such as testing, demonstration, analysis, and inspection, each of which should have a corresponding validation statement outlining how the requirement will be validated. By ensuring that requirements are clear and testable, validation can proceed concurrently with verification, allowing for practical implementation and integration into the system design. The process typically includes formal reviews, prototyping, and model checking to confirm that the requirements align with security goals and are technically feasible. This approach helps in identifying potential issues early,

preventing delays and avoiding costly redesigns, thereby ensuring that the system's security objectives are met efficiently.

#### 4.4.1 FORMAL REVIEWS

Formal reviews are structured evaluations of the requirements document, involving stakeholders such as cybersecurity experts, engineers, business owners, and end-users. The primary goal of these reviews is to assess the completeness, clarity, and feasibility of the requirements. During these sessions, participants scrutinize each requirement to ensure it accurately reflects the security needs and is realistically achievable. For CIE, formal reviews focus on assessing the alignment of requirements with CIE principles such as consequence-driven design, secure information architecture, and resilient layered defenses.

The feasibility of implementing CIE principles during formal reviews can be quantized by using scoring metrics. Each requirement can be rated on factors such as clarity, completeness, and alignment with CIE principles. For example, a requirement might be scored on a scale from 1 to 5 based on how well it specifies the necessary security measures. These scores can then be aggregated to provide an overall feasibility score for the requirements document.

#### 4.4.2 PROTOTYPING

Prototyping involves creating preliminary versions of the system or specific security features to validate their feasibility and effectiveness. In the context of CIE, prototypes are used to test the implementation of CIE principles such as active defense mechanisms, resilient layered defenses, and secure information architecture.

Prototyping also provides quantitative data on feasibility. Metrics such as the time required to implement specific security features, the performance impact of security measures, and the effectiveness of defenses under simulated attacks can be measured. These metrics provide a concrete basis for evaluating how feasible it is to implement CIE principles. For example, if a prototype reveals that a particular defense mechanism significantly degrades system performance, it might be deemed less feasible and require reconsideration.

#### 4.4.3 MODEL CHECKING

Model checking involves the use of formal methods to verify the correctness of security models against specified requirements. This technique is particularly useful for validating complex cybersecurity requirements that involve intricate interactions between system components. For CIE, model checking can be used to ensure that the requirements for secure information architecture, inter-dependency evaluation, and engineering information control are logically sound and implementable.

Model checking tools systematically explore all possible states of a system model to verify that it meets the specified security properties. For instance, a model checking tool might verify that access control policies are consistently enforced across all system states, or that data flow constraints prevent unauthorized data leakage. This rigorous analysis helps in identifying potential vulnerabilities and ensuring that the requirements are both necessary and sufficient to achieve the desired level of security.

The feasibility of implementing CIE principles through model checking can be quantized by assessing the complexity and resource requirements of the verification process. Metrics such as the number of states explored, the computational resources required, and the time taken for verification provide insights into the practical challenges of implementing the specified security measures. A high complexity or resource requirement might indicate that certain requirements are less feasible and need simplification or refinement.

#### 4.4.4 QUANTITATIVE FEASIBILITY ASSESSMENT

Quantifying the feasibility of implementing CIE principles involves measuring various aspects of the validation process and aggregating these measurements to provide a comprehensive feasibility assessment. Key quantifiable metrics include:

- **Clarity and Completeness Scores:** Derived from formal reviews, these scores assess how well requirements are specified and aligned with CIE principles.
- **Implementation Time and Performance Impact:** Obtained from prototyping, these metrics measure the practical effort and system performance implications of implementing specific security features.
- **Verification Complexity and Resource Requirements:** From model checking, these metrics indicate the computational and logistical feasibility of verifying complex security requirements.

By combining these metrics, organizations can create a feasibility index for CIE implementation. This index provides a numerical representation of how practical it is to implement the specified cybersecurity measures, enabling informed decision-making and prioritization.

In conclusion, the requirements validation stage for CIE is a multifaceted process that rigorously evaluates the feasibility of implementing CIE principles through formal reviews, prototyping, and model checking. By quantifying the outcomes of these activities, organizations can ensure that their cybersecurity requirements are not only theoretically sound but also practically achievable, leading to robust and resilient systems.

### 4.5. Requirements Management

When using CIE, the requirements management stage is essential to maintaining the integrity, consistency, and relevance of cybersecurity requirements throughout the system development life cycle. This stage encompasses several key activities, including change management, traceability, and version control, each tailored to address the dynamic nature of cybersecurity threats and the evolving needs of the system. Utilizing requirements management tools like DOORS [30], Innoslate [31], and Jama [32], organizations can effectively manage complex programs and ensure that cybersecurity measures remain robust, adaptable, and aligned with CIE principles, providing a resilient defense against emerging threats.

#### 4.5.1 CHANGE MANAGEMENT

Change management in the context of CIE is crucial, as cyber threats evolve rapidly, and new vulnerabilities can emerge with changing technologies and attack vectors. Requirements



management systems like DOORS facilitate the systematic process of evaluating, approving, and implementing changes to requirements by integrating change management workflows directly into the traceability framework. When new threats or system environment changes necessitate modifications to existing requirements, these tools help ensure that change proposals undergo thorough impact analysis. This analysis involves collaboration among cybersecurity experts, engineers, and business leaders to evaluate how the change affects the existing system and its security posture. The change is documented in a change request within the tool, reviewed by a change control board, and once approved, implemented with the requirements document updated accordingly. This systematic approach, supported by requirements management tools, ensures that changes are controlled, do not introduce new vulnerabilities, and are fully traceable.

#### 4.5.2 TRACEABILITY

Traceability is another critical aspect of CIE, ensuring that all cybersecurity requirements are addressed and linked to relevant design, implementation, and testing artifacts. Requirements management systems like DOORS and Jama provide robust traceability features that allow for creating and maintaining these links. For example, in CIE, a requirement for secure user authentication can be traced through the system from its origin to design specifications, implementation code, and test cases. These tools help identify gaps in the application of the CIE approach and can highlight where requirements might not map directly to CIE principles, which could indicate “gold plating” — exceeding the requirement rather than meeting it. Traceability matrices within these systems offer a visual representation of how requirements connect to other project artifacts, facilitating easy tracking, auditing, and compliance with regulatory standards. Moreover, they support impact analysis during change management, ensuring that necessary updates are consistently applied across the system, thereby minimizing the risk of introducing security gaps.

#### 4.5.3 VERSION CONTROL

Version control is equally critical in managing the evolution of requirements, particularly for complex systems with significant cybersecurity needs. Requirements management tools often include integrated version control systems that maintain a history of changes to the requirements document. These tools record each change with a unique version identifier, metadata, and a detailed history, which is invaluable for understanding the context and rationale behind each modification. This historical record is vital for decision-making, communication among stakeholders, and auditing and compliance. Furthermore, version control in requirements management tools supports parallel development efforts, allowing teams to work independently on different aspects of the system while maintaining a coherent overall project. For instance, a team might develop a new encryption feature based on updated requirements in a separate branch, later merging these changes back into the main project after thorough testing and validation. This capability is crucial for managing complex CIE-driven projects, where maintaining coherence and traceability across all changes is essential.

## 5. Conclusion

The Cyber Informed Engineering (CIE) requirements analysis framework serves as a structured approach to embedding cybersecurity considerations into the engineering lifecycle. By leveraging established engineering processes and integrating CIE principles, the framework provides a practical method for identifying, analyzing, and prioritizing requirements that address digital and operational risks.

This framework emphasizes the importance of consequence-driven design, engineering mitigations, and the alignment of cybersecurity measures with critical system functions. It ensures that requirements with direct or indirect digital consequences are identified and refined to minimize vulnerabilities and enhance resilience. Moreover, the framework promotes a collaborative effort among engineers, cybersecurity professionals, and other stakeholders, facilitating a comprehensive understanding of the security implications of system designs.

The CIE requirements analysis framework does not aim to replace existing methodologies but to augment them by encouraging a proactive, security-aware mindset throughout the engineering lifecycle. By adopting this framework, organizations can improve the quality of their requirements, prioritize critical security measures effectively, and develop systems that are robust against evolving cyber threats. This ensures that engineered systems not only meet functional goals but also maintain reliability and trustworthiness in an increasingly interconnected and digital world.

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NREL/TP-5T00-90317